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“It’s a digital world, but we always need an analogue bridge to it”

Analog Devices (ADI) offers signal processing and conditioning products. It caters to several markets including communication infrastructure, industrial, healthcare, automotive and consumer. ADI has close to \$3-billion turnover and 60,000 customers worldwide.

Somshubhro Pal Choudhury (Som), managing director-India, Analog Devices, spoke to Abhishek Mutha of EFY about the incremental versus fundamental change in research, the challenge faced while coming up with an advanced driver-assistance system, and why it is still an analogue world even though everything around us is digitised

In an almost wholly digital world, where do analogue components stand?

We live in a digital world but all of us interface to the physical world, which is analogue. Be it speech, image, temperature or radio signals, these are all inherently analogue. These analogue signals need to be captured, filtered or conditioned, amplified, and fed into the digital world using an analogue-to-digital converter, and finally processed digitally. Then the information goes back from this digital world to the analogue world. This is called the analogue signal chain. It would probably be apt to say it is a digital world but there is always an analogue bridge to the digital world.

Usually, while converting data from analogue to digital, we lose some information. Then why is conversion given so much importance?

You are absolutely right that we lose some information, but in the end it is all about precision. For example, while measuring temperature using a digital thermometer, the first question that arises is the precision required of the measurement—whether it is 98.4 or 98.45. An extra decimal point

signifies more accuracy and precision of the converters. Depending on the precision requirement, the right converter should be picked.

From a conversion standpoint, precision must be a big challenge. Are there any other challenges you would like to talk about?

Yes, there are several challenges while converting data from analogue to digital. Analogue signals are continuous signals, not discrete with respect to time, whereas digital signals are a distinct ‘0’ or ‘1.’ An analogue signal, being a continuous signal, gets highly impacted by the noise. The noise is contributed by the electronic circuitry where the analogue portion resides, thermal environment, interference and cross-talk even from other associated devices. The challenge is the conditioning of the incoming signal, i.e., eliminating or reducing the noise. In other words, it is signal filtering to minimise noise.

Low power is another challenge gaining importance from a system’s perspective. Here power refers to the amount of energy required by the converter to convert each bit of information.